Hardware components day 1

[Assingment Day 1]

[shaikh sohel]

1) Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



2) **Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).</em**

Feature	**Hard Disk Drive (HDD)**	**Solid State Drive (SSD)**
Speed	Relatively slow; restricted by mechanical rotation speeds (usually 5400 or 7200 RPM). Slower boot times and file transfers.	Significantly faster; uses flash memory for near-instant data access. Rapid boot times and high-speed file transfers.
Durability	Fragile; contains moving parts (spinning platters and moving heads) that are susceptible to damage from drops or physical shocks.	Highly durable; has no moving parts, making it resistant to physical vibration, shocks, and accidental drops.
Typical Use Cases	Bulk storage, data archiving, budget-friendly desktop PCs, and external backup drives where high capacity is needed at a low cost.	High-performance gaming PCs, laptops, operating system boot drives, professional video editing, and everyday multitasking.

3) Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

CPU (Central Processing Unit)

The CPU acts as the brain of your PC, managing core game logic, physics, artificial intelligence, and player inputs. It coordinates instructions between the game engine and other hardware components to ensure everything runs smoothly. Without a fast CPU, the game can stutter or lag during intense multiplayer matches.

RAM (Random Access Memory)

RAM provides fast, temporary storage for active game data, such as map layouts, player models, and textures. This allows the CPU to access information instantly without having to repeatedly load it from the slower storage drive. Having sufficient RAM prevents stuttering and ensures seamless transitions as you move across the battlefield.

GPU (Graphics Processing Unit)

The GPU is responsible for rendering the visual elements of the game, including 3D graphics, lighting, shadows, and textures. It processes millions of calculations simultaneously to produce high-resolution images and maintain a smooth frame rate on your monitor. A powerful GPU is crucial for delivering sharp visuals and responsive combat action.

4)Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

When upgrading a PC to run the latest titles like **EA Sports FC / FIFA** smoothly at target resolutions and frame rates (e.g., 1080p or 1440p at high refresh rates), consider upgrading the following key components:

Key Hardware Upgrades

Graphics Card (GPU)

Why: Modern FIFA titles rely heavily on advanced graphical engines for high-detail player models, dynamic lighting, realistic turf textures, and stadium crowd animations. Upgrading to a modern mid-range GPU (e.g., NVIDIA RTX 4060 / 5060 series or AMD Radeon RX 7600 series) ensures high-definition rendering and stable frame rates above 60 to 144+ FPS without graphical stutters.

Processor (CPU)

Why: EA Sports FC heavily stresses single-core processing for real-time match simulations, ball physics, collision detection, player AI, and stadium environment rendering. Upgrading to a newer mid-tier processor (such as an Intel Core i5 or AMD Ryzen 5/7 from recent generations) eliminates CPU bottlenecks, maintaining consistent frame times during chaotic dynamic plays.

System Memory (RAM)

Why: Latest titles require at least 8 GB–12 GB to run, but upgrading to **16 GB or 32 GB** of high-speed dual-channel RAM (DDR4 or DDR5) provides smooth operation. It prevents micro-stutters when

background apps (like Discord or browsers) run alongside the game and ensures faster asset retrieval during gameplay.

Solid-State Drive (SSD / NVMe)

Why: Traditional mechanical hard drives (HDDs) cause long boot times and delays loading into matches or Career Mode saves. Switching to an M.2 NVMe SSD dramatically speeds up game load times, menu navigation, asset streaming, and system responsiveness.